

Practical - 6

Practical Title: Rendering External JSON Data using Fetch API, Search and Filter

Problem Definition: Display event lists, student profiles, notices, or FAQs from external JSON files using Fetch API. Implement dynamic rendering, search, filter, sorting, and pagination.

A. Theory

1. How is JSON fetched, parsed, and rendered?

Answer: Fetched with `fetch()`, parsed using `response.json()`, and rendered by mapping data into HTML templates via `innerHTML`.

2. Which array methods are used for search, filter, sort, and pagination?

Answer: Used `filter()` and `includes()` for searching, `sort()` and `localeCompare()` for ordering, and `slice()` for pagination.

3. How are loading and error states handled?

Answer: Checked `response.ok` and used `.catch()` blocks to manage loading states and display fallback error messages.

4. How is modularity maintained in JavaScript files?

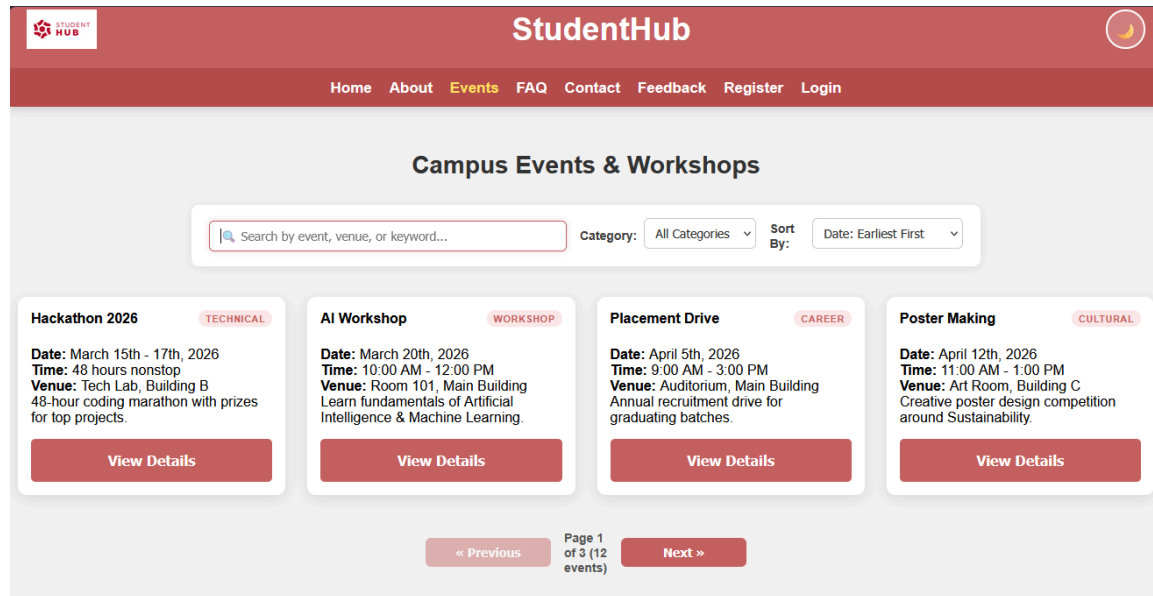
Answer: Encapsulated functionality into dedicated functions and files, separating data, business logic, and UI layers.

B. Implementation/ Output/Result

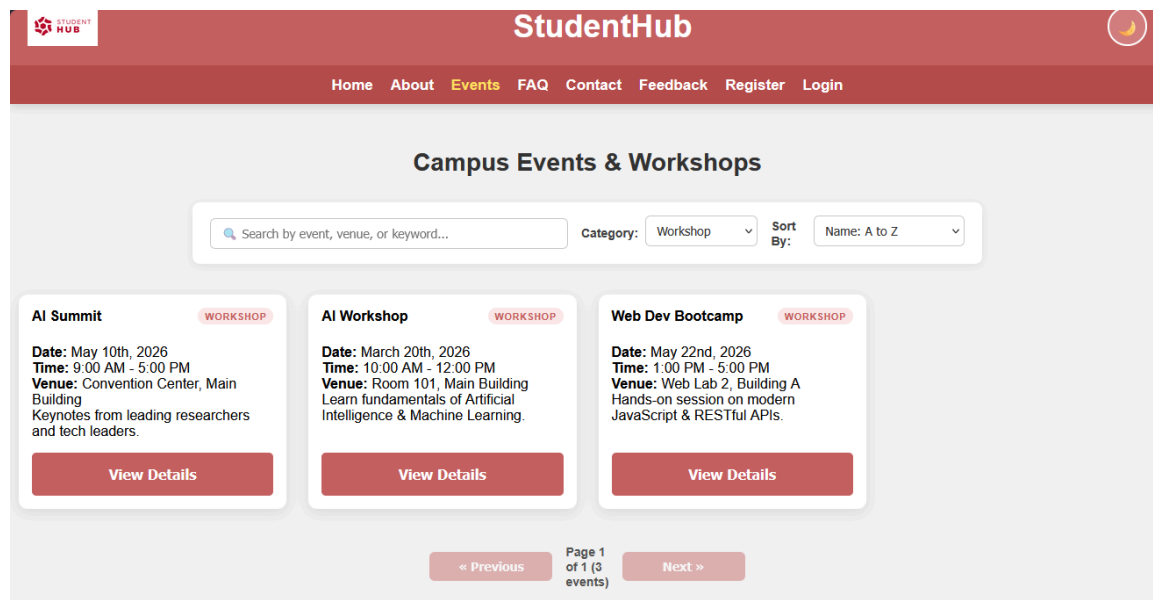
Feature	Description
JSON Data Store	Stores structured campus event records.
Fetch API and Rendering	Fetches JSON data asynchronously and renders cards dynamically.
Dynamic Search	Filters records in real time by title, venue and keyword.
Category Filter	Filters events by technical, workshop, career, or cultural type.

Sorting	Sort list by date (earliest/latest) or name (A-Z/Z-A)
Pagination	Breaks records into pages with Next/Previous navigation.

C. Evidence of Work (Screenshots)



Description: Figure 6.1 Dynamic rendering, searching, filtering, sorting in events page



Description: Figure 6.2 Filtered and sorted content in events page

D. GitHub Repository Link:

Link: <https://github.com/prep04-yl/StudentHub-Portal>

E. Challenges Faced:

Combining the search, filter and sorting simultaneously and pagination reset on filter or search.

F. Learning Outcome:

I learned how to work with external JSON data and render dynamic data and also implementing various features such as filter, search, and sort.

Faculty Remarks: (Keep this blank for faculty)